

GRANT WRITERS' SEMINARS AND WORKSHOPS

and the

**Arkansas Agricultural Experiment Station and Vice
Chancellor for Research & Innovation,
University of Arkansas and
Clinton School of Public Service**

present

WRITE WINNING GRANT PROPOSALS

**Tuesday, September 3, 2024 1:00-4:30 p.m. CDT; *and*
Wednesday, September 4, 2024 8:30 a.m.-12:00 p.m. CDT**

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University of Arkansas

Write Winning Grant Proposals

September 3-4, 2024

**Grant Writers' Seminars & Workshops
John D. Robertson, Ph.D.**

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GRANT WRITERS' SEMINARS AND WORKSHOPS

in association with

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WRITE WINNING GRANT PROPOSALS

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GRANT WRITERS' SEMINARS AND WORKSHOPS

Join the GWSW Proposal Writing Community

We invite you to join the GWSW proposal-writing community to receive (semimonthly) the latest funding agency updates, grant proposal writing tips and Q&A, information on how to write high-impact research manuscripts, and more.

EMAIL

CAPTCHA

☐

I'm not a robot



Submit

Check out GWSW's posted updates with information and helpful tips for preparing proposals to funding agencies and writing for biomedical publication. Access is provided to current entries as well as older relevant posts.

Visit URL www.grantcentral.com/blog/ to register for our E-list to receive periodic updates and insights for improving your grant proposal and manuscript writing skills.

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YOUR KEYS TO SUCCESS

- Your Idea!
- Your Commitment to the Process!
- Your Grantsmanship Skills!

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YOUR IDEA IS CRITICAL

HOW TO DEVELOP AN IRRESISTIBLE, FUNDABLE IDEA

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FOUR ELEMENTS TO DEVELOPING A GOOD IDEA FOR A GRANT

1. Be KNOWLEDGEABLE

- Learn everything you can about your field & identify a significant gap/lack/need in the field

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CRITICALLY ANALYZE RELATED LITERATURE

- Make use of reference databases, e.g.:
 - *Web of Science*
 - *PubMed*
 - *Google Scholar*
 - *Researcher (Mobile App)*

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FOUR ELEMENTS TO DEVELOPING A GOOD IDEA FOR A GRANT

1. Be KNOWLEDGEABLE

- Learn everything you can about your field & identify a significant gap/lack/need in the field

2. Be CREATIVE

- Devote time to thinking about a novel idea for a solution to the gap/lack/need

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UNDERSTANDING WHAT FUNDING AGENCIES FUND

For many federal (and some non-federal) funding agencies, it is possible to determine what the funding agency has funded in the near past, and what the funding agency is currently funding.

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UNDERSTANDING WHAT FUNDING AGENCIES FUND

Search databases of funded projects:

- NSF Awards
(<https://www.nsf.gov/awardsearch/advancedSearch.jsp>)
- USDA Enterprise Search
(<https://portal.nifa.usda.gov/enterprise-search/>)
- NIH RePORTER (Research Portfolio Online Reporting Tool)
(<https://reporter.nih.gov/advanced-search>)
- Institute of Education Sciences
(<https://ies.ed.gov/funding/grantsearch/index.asp>)
- DoD (CDMRP) Search Awards
(<https://cdmrp.army.mil/search.aspx>)

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ASSESS THE COMPETITION

THESE DATABASES CAN OFTEN BE
SEARCHED BY:

- Keywords
- Applicant
- Institution or state
- Funding mechanism
- Amount of the award
- Year of funding support
- Initial review panel/study section

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FOUR ELEMENTS TO DEVELOPING A GOOD IDEA FOR A GRANT

1. Be KNOWLEDGEABLE
 - Learn everything you can about your field & identify a significant gap/lack/need in the field
2. Be CREATIVE
 - Devote time to thinking about a novel idea for a solution to the gap/lack/need
3. Be OPEN
 - Share your ideas with knowledgeable colleagues
4. Be ADAPTABLE
 - Refine the problem and/or solution to maximize impact on the field

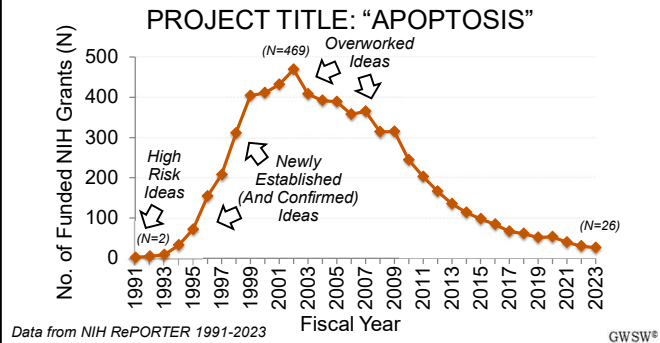
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WHAT DO FUNDING AGENICES FUND / NOT FUND

- High risk ideas/concepts are very difficult to get funded
- Newly established (and confirmed) ideas/concepts are frequently funded
- Extensively pursued or “well worn” ideas/concepts are more difficult to get funded

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GRANT WRITERS' SEMINARS AND WORKSHOPS



EVALUATION OF YOUR IDEA ~AGENCY PRIORITIES~

Even if your idea is novel, not all good ideas are necessarily good ideas from the framework of funding agency priorities. Thus, it is critical that you assess early on the programmatic relevance of your idea.

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ENSURE THE PROGRAMMATIC RELEVANCE OF YOUR IDEA

CONTACT NIH PROGRAM OFFICERS

"NIH staff is here to help. We **strongly encourage** applicants and grantees to communicate with us throughout the grant life cycle. Understanding the roles of NIH staff can help you contact the right person at each phase of the application and award process."

<https://grants.nih.gov/grants/how-to-apply-application-guide.html>
General Instructions for NIH and Other PHS Agencies, SF424
(R&R) Application Packages, G.110, Application Process

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ENSURE THE PROGRAMMATIC RELEVANCE OF YOUR IDEA

CONTACT NSF PROGRAM OFFICERS

"The NSF website provides the most comprehensive source of information on NSF Directorates (including *contact information*), programs and funding opportunities. Use of this website by potential proposers is *strongly encouraged*."

<https://new.nsf.gov/policies/papgg/24-1>
NSF Proposal and Award Policies and Procedures Guide,
Chapter 1B. NSF Programs and Funding Opportunities

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ENSURE THE PROGRAMMATIC RELEVANCE OF YOUR IDEA

NIFA PROGRAM AREA PRIORITY CONTACTS

"After reading the RFA, applicants *often will contact the program staff to discuss the applicability of a topic to the program goals* and suitability for prospective submission as an application. Applicants are encouraged to submit only those applications responsive to the funding priorities outlined in the RFA."

The NIFA Peer Review Process for Competitive Grant Applications
<https://www.nifa.usda.gov/nifa-peer-review-process-competitive-grant-applications>

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PROGRAM OFFICERS ARE OF GREAT VALUE TO YOU

Planning Phase:

- Help choose funding mechanism and advise on needed qualifications

Writing Phase:

- Can help *"maximize programmatic relevance"*

Submission:

- Provide insights regarding reviewers

Review Phase:

- Feedback on proposal's review and on future submission

Funding Phase:

- Advocate within a given program

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ENSURE THE PROGRAMMATIC RELEVANCE OF YOUR IDEA

CONTACT NEH PROGRAM OFFICERS

"They can provide information about the division's application guidelines and the eligibility or competitiveness of potential project ideas, and provide tips about common proposal-writing mistakes to avoid. They will supply samples of successful application narratives in each grant category and *even provide feedback for a preliminary draft of a proposal if it is submitted well before the deadline*. All potential applicants are *encouraged* to contact a program officer early in their project conceptualization process."

<https://www.neh.gov/divisions/public/>
NEH Division of Public Programs GWSW®

CONTACTING PROGRAM OFFICERS

- Do not contact a Program Officer until you are truly prepared to have a meaningful discussion about your ideas (re: *how to maximize programmatic relevance*).
- Always remember that Program Officers are human beings and will reflect a range of personalities. They may, or may not, be particularly helpful to you.

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ASSESS YOURSELF

After developing a readily fundable idea, critically assess whether you have the necessary *expertise, resources, personnel, and / or preliminary data* to be competitive.

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COMMITMENT TO THE PROCESS

- To keep current on the literature
- To find funding opportunities
- To write the application
- To get critical review from knowledgeable colleagues
- To accept and respond *positively* to criticism
- To do whatever it takes

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COMMITMENT TO THE PROCESS

Arguably, the *single most important* problem in successful grantsmanship is the lack of *commitment* to find/make the time necessary to prepare and write a truly competitive proposal.

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COMMON REASONS FOR APPLICATION FAILURE

1. Problem to be studied is not clearly stated
2. Lack of rationale for the project
3. Insufficient knowledge of the literature
4. Lack of essential experience of applicant
5. Diffuse, superficial or unfocused approach
6. Applicant failed to follow instructions
7. Unrealistic amount of work proposed
8. Uncertain outcomes and future directions

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COMMON REASONS FOR APPLICATION FAILURE

9. Unrealistic budget
10. Not relevant to mission of funding agency
11. Interdependence of aims/goals
12. Reader-unfriendly application
13. Misinterpreted deadline for application
14. Applicant did not address review criteria
15. Insufficient preliminary data
16. Etc., etc., and etc.

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COMMON REASONS FOR APPLICATION FAILURE

**LACK OF
GRANTSMANSHIP
SKILLS!**

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THE IMPORTANCE OF GRANTSMANSHIP SKILLS

Successful Grantsmanship requires that you pay attention to:

- Mechanics – know how to fill out the application forms *correctly*
- Concepts – understand the *purpose* of grant applications
- Psychology – appreciate that grant applications are read by *people*
- Logic – understand that your idea must be *understandable* to be successful

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THE BOTTOM LINE

It is critical that you sell your idea to the consumer group (reviewers). Therefore, you must *make a good first impression* and be *credible in your identification of a need*. To do so requires that you are *well prepared*. Then, you need to present to reviewers your *novel idea for a solution* (i.e., "the product") in a way that is *maximally understandable*. This should position your idea so that it is clear you have *something special to offer*.

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YOUR KEYS TO SUCCESS

- Your Idea!
- Your Commitment to the Process!
- Your Grantsmanship Skills!

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GRANT WRITERS' SEMINARS AND WORKSHOPS

*FUNDING
OPPORTUNITIES*

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FEDERAL GRANT FUNDING OPPORTUNITIES

- National Institutes of Health
- National Science Foundation
- National Endowment for the Arts
- National Endowment for the Humanities
- U.S. Department of Agriculture (NIFA)
- Center for Disease Control/Prevention
- National Institute of Justice
- National Inst. of Occupational Safety & Health
- Department of Education
- Department of Defense
- Department of Homeland Security and others

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NATIONAL ORGANIZATION GRANT FUNDING

- American Heart Association
- American Cancer Society
- American Lung Association
- Arthritis Foundation
- American Association for Colleges of Pharmacy
- American Society of Plant Biologists
- National Board of Medical Examiners
- National Organization for Rare Disorders
- National Art Education Association
- American Theater Wing (National Theater Company Grants)

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FOUNDATION GRANT SUPPORT

THERE ARE LITERALLY THOUSANDS OF FOUNDATIONS

“Since 90% of U.S. foundations do not have websites, the insights you need to find funders will not be discovered via standard search engines.”*

*Foundation Directory Online
<https://fconline.foundationcenter.org/>

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IRS FORM 990

“Form 990 provides information on *grants and other assistance* made by the filing organization during the tax year to domestic organizations, domestic governments, and domestic individuals.”

<https://candid.org/research-and-verify-nonprofits/990-finder>

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SEARCH THE INTERNET

THE FOUNDATION CENTER

<https://fconline.foundationcenter.org/>

PIVOT-RP

<https://pivot.proquest.com/>

GOOGLE

<https://www.google.com>

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FOUNDATION GRANT SUPPORT

ALL ARE DIFFERENT!

*“When you’ve seen one foundation –
you’ve seen one foundation.”*

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FOUNDATION GRANT SUPPORT

- Amount of award is generally much less
- Tenure of award is usually shorter
- Indirect costs will be less (or none)
- Mission of Foundation usually highly restricted
- Application format is shorter
- Feedback on unsuccessful proposals is uncommon
- Personal relationships can be helpful

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KNOW HOW APPLICATIONS ARE MADE TO FOUNDATIONS

- Talk to a Foundation Program Officer (if allowed)
- Check out the Foundation's Website to identify funding opportunities
- Track recently funded grants by reviewing the Foundation's annual reports
- Be certain you understand the Foundation's *Mission Statement/Priorities*

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GRANT WRITERS' SEMINARS AND WORKSHOPS

FORTUNATELY.....

GRANT APPLICATIONS TO **VERY DIFFERENT** FUNDING AGENCIES ARE **VERY SIMILAR** !

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
Overview	X	X	X	X	X	X
Significance	X	X	X	X	X	X
Budget	X	X	X	X	X	X
Applicant/Env't	X	X	X	X	X	X
Background	X	X	X	X	X	X
Previous experience or preliminary data	X	X	X	X	X	X
Plan of work	X	X	X	X	X	X

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GRANT WRITERS' SEMINARS AND WORKSHOPS

PREPARATION OF THE APPLICATION

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
Significance	X	X	X	X	X	X
Budget	X	X	X	X	X	X
Applicant/Env't	X	X	X	X	X	X
Background	X	X	X	X	X	X
Previous experience or preliminary data	X	X	X	X	X	X
Plan of work	X	X	X	X	X	X

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EVERY GRANT PROPOSAL SHOULD INCLUDE AN OVERVIEW SECTION

THE “EXECUTIVE SUMMARY” OF YOUR APPLICATION

- NSF: First part of Project Description entitled “Overview and Objectives” section
- USDA: “Overview” of “Introduction” section + “Rationale” of “Rationale & Significance” section
- NIH: “Specific Aims” section
- NEH: First part of “Significance and Contribution” section of the Narrative
- ACS: Specific Aims page
- DoD: Specific Aims page
- NASA: “Overview and Significance” section

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OVERVIEW SECTION

This section **MUST** contain everything that is important and exciting about your project, but without a lot of detail.

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OVERVIEW SECTION

- This is one of the two *most important* sections in any grant application
- It is by far the most difficult section of your application to write well
- It **MUST** quickly engender robust enthusiasm for your idea
- The flow of logic must be compelling
- It serves as the template for the rest of your research plan or project narrative/description
- Write the Overview section of your proposal **FIRST**

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OVERVIEW SECTION

In most cases, this is the point of entry for the reviewer to the main body of your research proposal. Consequently, it is highly probable that s/he will come to a general conclusion about you, the importance of your ideas, and the clarity of your thinking by the time s/he finishes reading this short "Overview" section of your proposal.

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OVERVIEW SECTION

STRUCTURE OF THE OVERVIEW

- 1st paragraph: Identify the "Need"
- 2nd paragraph: Outline the Idea for a Solution
- 3rd "paragraph": Spell out the Approach
- 4th paragraph: Summarize Expectations – (Payoff to the Funding Agency)

ALL ON ABOUT ONE PAGE

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EXAMPLE #1: OVERVIEW SECTION

Example: You are a social scientist who has an interest in decreasing health disparities related to tobacco control. Your pilot data suggest that nearly 90% of underserved mothers who quit smoking during pregnancy have resumed smoking by the time infants are only three months old. Your pilot data also suggest that psychosocial and environmental factors may play a role in smoking relapse. Recently, a NOFO was issued seeking research investigating health disparities among minority and underserved women of child-bearing age. You decide to respond to this funding opportunity.

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EXAMPLE #2: OVERVIEW SECTION

Example: You are a civil engineer with expertise in concrete and structures. An unresolved problem in the field relates to the inability to nondestructively detect delamination-type defects after concrete has been poured. You have an idea, based in part upon pilot data, to use ground penetrating radar and impact echo techniques to detect such defects. Recently, a NOFO was issued seeking research focusing on novel approaches to detect and/or repair internal defects in concrete bridge decks. You and your team decide to respond to this funding opportunity.

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THE COMPONENTS MUST CREATE A LINEAR PROGRESSION OF LOGIC

Introductory Paragraph

Opening sentence
Current knowledge
Gap or lack
Statement of need
Consequences of not meeting need

Specifics "Paragraph"

Specific objectives/aims

Payoff Paragraph

Expected outcomes
Positive impact

What, Why, Who Paragraph

Long-term goal
Overall objective in this application
Central hypothesis & how formulated
Rationale
Why well prepared

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GRANT WRITERS' SEMINARS AND WORKSHOPS

CREATE A BULLETED OUTLINE

See Appendix One to this handout for example and/or your *Workbook* for further information

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OVERVIEW SECTION

INTRODUCTORY PARAGRAPH

The primary purpose of the opening paragraph is to convince all reviewers (through a process of *education*) that there is a significant unknown/issue/debate (i.e., a “*problem*”). The *problem* then provides the basis for the “critical need” relevant to the mission of the funding agency (e.g., NIH, NSF, DoJ, USDA, etc.).

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OVERVIEW SECTION

OPENING SENTENCE

- Use a “hook” sentence to begin the flow of logic necessary to *recruit* the reviewer to become an advocate for your grant proposal.
- Must address two key points:
 - *First*, it should generally identify what the proposal will be about; and
 - *Second*, it should immediately relate to the mission of the funding agency.
- It should not simply state knowledge that will be obvious to any reviewer

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OVERVIEW SECTION

OPENING SENTENCE

Unhelpful example:

Cigarette smoking is recognized to cause adverse effects on health.

Helpful example:

Although overall rates of cigarette smoking in the U.S. have decreased during the past 20 years, smoking rates have increased among underserved women of child-bearing age.

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OVERVIEW SECTION

OPENING SENTENCE

Unhelpful example:

The long-term stability of reinforced concrete structures is critical for safety of America's highway bridges.

Helpful example:

Detection of delamination defects in reinforced concrete slabs is an essential, but currently difficult, aspect of effective construction and maintenance of highway bridges.

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OVERVIEW SECTION

CURRENT KNOWLEDGE

- Must *educate* the reviewer with what is important in the field identified in the opening sentence
- Should be designed to, in 3-5 pithy sentences, bring the reviewer up to speed with the "state-of-the-art" in the field
- Cite only key references here
- All key points that reviewers need to know **MUST** be introduced in this first paragraph
- *Should provide the conceptual framework for the presentation of the.....*

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OVERVIEW SECTION

FRAMING OF THE PROBLEM

- The body of "important knowns" must lead to the conclusion that there is an *important* gap in knowledge or lack of something. Without a "*gap/lack*," there's *no basis for the development* of a grant proposal
- The "gap/lack" **MUST**, in turn, make the case for a real "problem" and the argument that there is.....
- A *critical need* or *urgent need*

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EXAMPLE #1

FRAMING OF THE PROBLEM

What remains unknown are predictors of smoking relapse in this population. Thus, there is a *critical need* to identify a list of factors associated with cigarette smoking recidivism among underserved new mothers shortly after giving birth.

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EXAMPLE #2

FRAMING OF THE PROBLEM

The combined evidence supports that nondestructive testing methods for detection of such flaws in reinforced concrete slabs remain inadequate. Thus, there is a *critical need* to design improved noninvasive methods to detect delamination defects in reinforced concrete structures.

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OVERVIEW SECTION

HOW *NOT* TO FRAME A PROBLEM

- "Little research has been done..... Therefore, there is a need for more research....."
- "No one has studied the effect of..... There is, therefore, a need to study....."
- "Relatively few studies have been made on....."
- "To the best of our knowledge, there is currently no information available on....."
- "No publications have examined the reasons responsible for....."

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OVERVIEW SECTION

FRAMING OF THE PROBLEM

- The body of “important knowns” must lead to the conclusion that there is an *important* gap in knowledge or lack of something. Without a “gap/lack,” there’s *no basis for the development* of a grant proposal
- The “gap/lack” MUST, in turn, make the case for a real “problem” and the argument that there is.....
- A *critical need* or *urgent need*
- *Should conclude with why the absence of a “solution” to this compelling problem is an issue for the funding agency—NOT your idea for a solution to the critical need*

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EXAMPLE #1

CONSEQUENCE OF NOT MEETING NEED

Thus, there is a *critical need* to identify a list of factors associated with cigarette smoking recidivism among underserved new mothers shortly after giving birth. *In the absence of such knowledge, the development of effective intervention strategies to help decrease smoking-related health risks within this at-risk population will likely remain difficult.*

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EXAMPLE #2

CONSEQUENCE OF NOT MEETING NEED

Thus, there is a *critical need* to design improved noninvasive methods to detect delamination flaws in reinforced concrete structures. *In the absence of such methods, the promise of extending the service life of U.S. highway bridges through improved construction and maintenance will likely remain difficult.*

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OVERVIEW SECTION

CRITICAL POINT!

If, by the end of the opening paragraph, you have not convinced every reviewer of your application that there is a *bona fide* significant “problem” or “need” (that is relevant to the *mission* of the funding agency), then unfortunately everything that follows (i.e., your idea) will be very difficult to “sell” to reviewers!

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THE COMPONENTS MUST CREATE A LINEAR PROGRESSION OF LOGIC

Introductory Paragraph

Opening sentence
Current knowledge
Gap or lack
Statement of need
Consequences of not meeting need

Specifics “Paragraph”

Specific objectives/aims

Payoff Paragraph

Expected outcomes
Positive impact

What, Why, Who Paragraph

Long-term goal
Overall objective in this application
Central hypothesis & how formulated
Rationale
Why well prepared

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OVERVIEW SECTION

SECOND PARAGRAPH

The primary purpose of the second paragraph is to convince all reviewers that you and the research team (i.e., your colleagues) have the “Solution” to the “Problem” or issue (i.e., a *fix* for the critical need) identified in the introductory paragraph.

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OVERVIEW SECTION

LONG-TERM GOAL

- This is not the goal of the current application; instead, this is your career goal (of which the current application will be only one part)
- By definition, your long-term goal and the mission of the funding agency are one
- Be realistic: do not overstate or over-anticipate your capabilities

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OVERVIEW SECTION

LONG-TERM GOAL

Unhelpful example:

The long-term goal is to eliminate illness, disability, and death related to tobacco use.

Helpful example:

The long-term goal is to identify key factors predictive of tobacco use, so that improved strategies can be developed to reduce environmental tobacco smoke exposure and tobacco-related morbidity and mortality.

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OVERVIEW SECTION

LONG-TERM GOAL

Unhelpful example:

Our long-term goal is to design and construct all highways and bridges with optimal safety features.

Helpful example:

Our long-term goal is to establish effective strategies for design and construction of reinforced concrete structures for major building projects with optimal safety features.

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OVERVIEW SECTION

OVERALL OBJECTIVE

- This is where the purpose of the project is identified (Not in the 1st paragraph or long-term goal)
- Should define the overall purpose of the proposed set of activities, experiments, etc.
- Must be crafted in such a way that it addresses the need that was identified in the 1st paragraph
- Must be appreciated as a step toward attainment of the long-term goal
- Must always have a well-defined endpoint

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OVERVIEW SECTION

OVERALL OBJECTIVE

Unhelpful example:

Our overall objective in this application is to investigate factors related to high smoking relapse rates among underserved new mothers.

Why is this an unhelpful objective?

1. It is **indeterminate in endpoint** (i.e., “investigate”)
2. It is a “look-to-see” objective

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OVERALL OBJECTIVE

INDETERMINATE OBJECTIVES

-to investigate factors related to.....
-to explore why.....
-to better understand why.....
-to improve our understanding of.....
-to examine the reasons for.....
-to research the effects of.....
-to study the mechanisms of.....

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OVERVIEW SECTION

OVERALL OBJECTIVE

Unhelpful example:

Our overall objective in this application is to *look to see what* factors *are* related to high smoking relapse rates among underserved new mothers.

Why is this an unhelpful objective?

1. It is indeterminate in endpoint (i.e., “investigate”)
2. It is a “look-to-see” objective

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OVERVIEW SECTION

OVERALL OBJECTIVE

Unhelpful example:

Our overall objective in this application is to investigate factors related to high smoking relapse rates among underserved new mothers.

Helpful example:

Our overall objective in this application is to identify factors predictive of high smoking relapse rates among underserved new mothers.

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OVERVIEW SECTION

OVERALL OBJECTIVE

Unhelpful objective:

Our overall objective in this application is to explore whether ground penetrating radar and impact echo techniques can be used to detect delamination defects in reinforced concrete.

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OVERVIEW SECTION

OVERALL OBJECTIVE

Unhelpful objective:

Our overall objective in this application is to *look to see* whether ground penetrating radar and impact echo techniques can be used to detect delamination defects in reinforced concrete.

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OVERVIEW SECTION

OVERALL OBJECTIVE

Unhelpful objective:

Our overall objective in this application is to explore whether ground penetrating radar and impact echo techniques can be used to detect delamination defects in reinforced concrete.

Helpful objective:

Our overall objective in this application is to design improved non-destructive diagnostic procedures for detection of delamination flaws in reinforced concrete.

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OVERVIEW SECTION

After you have formulated your statement of the overall objective in this application, check to make certain that it closely matches the identified *critical need* and that it represents a step toward the attainment of the *long-term goal*. If it does not, you have a problem.

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OVERVIEW SECTION

ATTAINING YOUR OVERALL OBJECTIVE:

- *Formulate a “central hypothesis”*
Or
- Directly address the “need”

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OVERVIEW SECTION

CENTRAL HYPOTHESIS

Make certain that you write a real hypothesis:

A tentative assumption made in order to draw out and test its logical or empirical consequences

Merriam-Webster

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TIPS ON WRITING YOUR HYPOTHESIS

- A proposal with a central hypothesis requires the existence of a certain body of evidence related to the stated overall objective
- The central hypothesis would be your “best bet” for an outcome among all alternatives
- It should have readily identifiable parts that, in turn, set up writing of the specific aims
- Implicit in any hypothesis is that it could be invalid
- To assess its validity, it must be testable; thus, there must always be at least two outcomes

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CENTRAL HYPOTHESIS

The difference between a testable and non-testable (non-committal) hypothesis:

Incorrect: Our central hypothesis is that key psychosocial and environmental factors *may* relate to the high rates of smoking recidivism among at-risk new mothers.

Correct: Our central hypothesis is that key psychosocial and environmental factors relate to the high rates of smoking recidivism among at-risk new mothers.

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CENTRAL HYPOTHESIS

The difference between a testable and non-testable (predetermined outcome) hypothesis:

Incorrect: Our central hypothesis is to *prove* that key psychosocial and environmental factors relate to the high rates of smoking recidivism among at-risk new mothers.

Correct: Our central hypothesis is that key psychosocial and factors relate to the high rates of smoking recidivism among at-risk new mothers.

GWSW®

LINKAGE OF THREE KEY COMPONENTS

1. Long-term goal: *Broadest*

- Projects an obvious progression of career-based research activities

2. Overall objective in this application: *Narrower*

- Current step along the progression of research activities that will achieve the “critical need” identified

3. Central hypothesis: *Narrowest*

- Best bet among the possible outcomes; MUST be testable and therefore potentially invalid

GWSW®

LINKAGE OF THREE KEY COMPONENTS

1. *Long-term goal: Broadest*

The long-term goal is to identify important factors predictive of tobacco use, so that improved strategies can be developed to reduce environmental tobacco smoke exposure and tobacco-related morbidity and mortality.

2. *Overall objective: Narrower*

Our overall objective in this application is to identify factors predictive of high smoking relapse rates among underserved new mothers.

3. *Central hypothesis: Narrowest*

Our central hypothesis is that key psychosocial and environmental factors relate to the high rates of smoking recidivism among at-risk mothers.

GWSW*

LINKAGE OF THREE KEY COMPONENTS

1. *Long-range goal: Broadest*

The long-range goal is to establish effective strategies for design and construction of reinforced concrete structures for major building projects with optimal safety features.

2. *Overall objective: Narrower*

The overall objective in this application is to design improved nondestructive diagnostic procedures to detect delamination flaws in reinforced concrete.

3. *Central hypothesis: Narrowest*

Our central hypothesis is that application of ground penetrating radar and impact echo techniques will allow accurate nondestructive detection of delamination defects in reinforced concrete structures.

GWSW*

CENTRAL HYPOTHESIS

- Since hypotheses are, by definition, based upon an existing body of evidence, you as the applicant have an obligation to provide that evidence
- Our hypothesis has been formulated, in large part, based on our own preliminary findings demonstrating that...

GWSW*

OVERVIEW SECTION

ATTAINING YOUR OVERALL OBJECTIVE:

- Formulate a “central hypothesis”
Or
- *Directly address the “need”*

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NEEDS-BASED APPLICATION

Example: You are the same social scientist who has an interest in decreasing health disparities related to tobacco control. In your initial studies, you find that nearly 90% of low-income mothers who quit smoking during pregnancy have resumed smoking by the time infants are only three months old. However, at the time you are writing your proposal, you have no idea about why (i.e., very few preliminary data). You decide to go for a seed / pilot data grant.

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LINKAGE OF THREE KEY COMPONENTS

1. *Long-term goal: Broadest*

The long-term goal is to identify key factors predictive of tobacco use, so that improved strategies can be developed to reduce environmental tobacco smoke exposure and tobacco-related morbidity and mortality.

2. *Overall objective: Narrower*

Our overall objective in this application is to identify factors predictive of high smoking relapse rates among low-income new mothers.

3. *Rationale: Justification of need*

The rationale is that new pilot evidence on predictors of high smoking recidivism in this underserved population would provide focus for, and supportive preliminary data of, subsequent definitive studies.

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OVERVIEW SECTION

RATIONALE

The underlying reason: BASIS

Merriam-Webster

- Every Overview section should have a statement of rationale
- Ensure that you formulate a true rationale
- What is likely to become possible after the project is finished that is not possible now
- The rationale is the underlying reason you decided to pursue the project in the first place
- The statement of rationale should relate well to the critical need and the mission of the funding agency

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OVERVIEW SECTION

APPLICANT(S)/ORGANIZATION(S): WHY MOST QUALIFIED

- Your opportunity to be *proactive* regarding your collective qualifications
- *Not* routine biographical information
- Summary of *collective reasons* why you have the competitive advantage, e.g.:
 - *Unique* qualifications of work team
 - Quality & quantity of preliminary data
 - *Unique* skills, technologies, past successes

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OVERVIEW SECTION

WHY WELL-POSITIONED

Unhelpful example:

"I am well prepared to undertake this project because of my passion for decreasing health disparities."

Helpful example:

"I am well prepared to undertake this project based on my previous publications on tobacco-related health disparities in urban settings, and I have access to random, stratified, monthly survey data of recent mothers of low socioeconomic status selected from birth certificates (see letter of support)."

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THE COMPONENTS MUST CREATE A LINEAR PROGRESSION OF LOGIC

Introductory Paragraph

Opening sentence
Current knowledge
Gap or lack
Statement of need
Consequences of not meeting need

Specifics "Paragraph"

Specific objectives/aims

Payoff Paragraph

Expected outcomes
Positive impact

What, Why, Who Paragraph

Long-term goal
Overall objective in this application
Central hypothesis & how formulated
Rationale
Why well prepared

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OVERVIEW SECTION

THIRD "PARAGRAPH"

The primary purpose of the third "paragraph" is to provide a logical step-by-step development of the specific activities (Aims/Objectives) whereby you will fulfill the identified overall objective and/or test the central hypothesis to completely address the "critical need."

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OVERVIEW SECTION

FORMULATION OF SPECIFIC AIMS/OBJECTIVES

- Two to four concise, eye-catching "headline" statements
- If possible, conceptual or outcome focused, NOT descriptions of *activities*
- Each should be focused by a subordinate *working hypothesis* or *approach* statement

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AN UNFOCUSED SPECIFIC AIM

Specific aim #1: Study psychosocial factors of underserved mothers who resume smoking after giving birth. We hypothesize, based upon pilot data, that psychosocial factors will prove to be an important variable that will predict smoking relapse.

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A DESCRIPTIVE SPECIFIC AIM

Specific aim #1: Investigate whether psychosocial factors play a role in smoking relapse among underserved new mothers. We hypothesize that this will be an important predictive factor for relapse.

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A DESCRIPTIVE SPECIFIC AIM

Specific aim #1: Look to see whether (or not) psychosocial factors play a role in smoking relapse among underserved new mothers. We hypothesize that this will be an important predictive factor for relapse.

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A CONCEPTUAL SPECIFIC AIM

Specific aim #1: Identify key predictive psychosocial factors for smoking relapse among underserved new mothers. Our working hypothesis is that loneliness, perceived infant irritability, and / or decreased self image will predict smoking relapse among at-risk new mothers.

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A DESCRIPTIVE SPECIFIC OBJECTIVE

Specific objective #1: Explore whether ground penetrating radar can be used to detect deterioration defects in concrete. Our working hypothesis is that this technique will be effective up to depths of one meter from the surface.

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A DESCRIPTIVE SPECIFIC OBJECTIVE

Specific objective #1: Look to see whether (or not) ground penetrating radar can be used to detect deterioration defects in concrete. Our working hypothesis is that this technique will be effective up to depths of one meter from the surface.

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A CONCEPTUAL SPECIFIC OBJECTIVE

Specific objective #1: Identify a technique for rapid detection of general subsurface deterioration in reinforced concrete. The working hypothesis is that ground penetrating radar will provide deterioration mapping at depths up to one meter from the surface.

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SPECIFIC AIMS/OBJECTIVES

Each of the aims/objectives should be related to the other stated aims/objectives but avoid having the feasibility of one aim/objective depend upon a particular outcome of another aim/objective.

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SPECIFIC AIMS/OBJECTIVES

Tie the aims together but avoid having the feasibility of one aim depend on a particular outcome of another aim.

HYPOTHESIS: Our central hypothesis is that application of ground penetrating radar and impact echo techniques will allow accurate non-destructive detection of delamination in reinforced concrete structures.

OBJ 1. Identify a technique for rapid detection of general subsurface deterioration in reinforced concrete....

OBJ 2. Examine whether impact echo techniques will identify delamination in zones of deterioration detected by ground penetrating radar.

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SPECIFIC AIMS/OBJECTIVES

Tie the aims together but avoid having the feasibility of one aim depend on a particular outcome of another aim.

HYPOTHESIS: Our central hypothesis is that application of ground penetrating radar and impact echo techniques will allow accurate non-destructive detection of delamination in reinforced concrete structures.

OBJ 1. *Identify a technique for rapid detection of general subsurface deterioration in reinforced concrete....*

OBJ 2. *Look to see whether (or not) impact echo techniques will identify delamination in zones of deterioration detected by ground penetrating radar.*

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SPECIFIC AIMS/OBJECTIVES

HYPOTHESIS: Our central hypothesis that application of ground penetrating radar and impact echo techniques will allow accurate non-destructive detection of flaws in reinforced concrete structures.

OBJ 1. *Identify a technique for rapid detection of general subsurface deterioration in reinforced concrete. Answer: Ground penetrating radar was not able to detect defects.*

OBJ 2. *Examine whether impact echo techniques will identify delamination in zones of deterioration detected by ground penetrating radar.*

Answer: There is no signal for characterization!!!

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SPECIFIC AIMS/OBJECTIVES

OBJ 1. *Identify a technique for rapid detection of general subsurface deterioration in reinforced concrete.* The working hypothesis is that ground penetrating radar will provide deterioration mapping at depths up to one meter from the surface.

OBJ 2. *Determine the effectiveness of the impact echo method for identifying delamination in deteriorated reinforced concrete.* The working hypothesis is that impact echo methods will provide sensitive and unambiguous detection and characterization of delamination flaws in reinforced concrete.

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THE COMPONENTS MUST CREATE A LINEAR PROGRESSION OF LOGIC

Introductory Paragraph

Opening sentence
Current knowledge
Gap or lack
Statement of need
Consequences of not meeting need

Specifics "Paragraph"

Specific objectives/aims

Payoff Paragraph

Expected outcomes
Positive impact

What, Why, Who Paragraph

Long-term goal
Overall objective in this application
Central hypothesis & how formulated
Rationale
Why well prepared

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OVERVIEW SECTION

FOURTH (FINAL) PARAGRAPH

The primary purpose of the fourth paragraph is to inform the reviewers (and the funding agency) exactly what the "return on the investment" (the deliverables) will be and why this will be of value to the mission of the funding agency.

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OVERVIEW SECTION

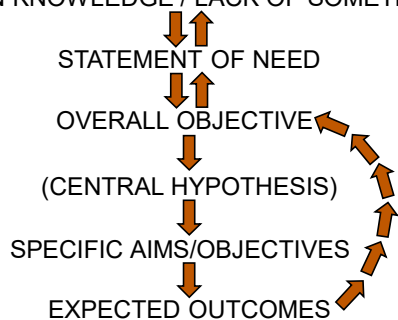
EXPECTATIONS AND IMPACT

- This paragraph should include expected outcomes, which must be specific and credible: this is the "return on the sponsor's investment" in you and your project
- Do not write this in the future tense; this is what **will have** been accomplished (i.e., We expect to have determined...)
- Conclude paragraph with positive impact (i.e., a general statement of how these outcomes will fill the identified need and thereby advance the mission of the agency)

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LINEAR PROGRESSION FOR A STRONG OVERVIEW SECTION

GAP IN KNOWLEDGE / LACK OF SOMETHING



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GRANT WRITERS' SEMINARS AND WORKSHOPS

See Appendix Two to this handout for example of an Overview Section and/or your *Workbook* for further information

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OVERVIEW SECTION

THE BOTTOM LINE!

Independent of what funding agency you ultimately will be targeting, or what the specific format that funding agency requires, you will *always* be well served by preparing an Overview section (or equivalent) of your proposed research plan *as a critical first step in the development of any grant application*.

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THE REMAINDER OF THE PROPOSAL

If you have conscientiously developed your grant proposal to this point (and more specifically, your Overview section), preparation of the rest of the proposal should be easy, provided that...

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CHECKLIST FOR WRITING THE PROPOSAL

WHAT SHOULD BE IN PLACE:

- A critically peer-reviewed and polished Overview section of your proposal
- One or more discussions with an appropriate Program Officer
- An intimate familiarity with the idea / field on which the proposal will be based
- Sufficient time identified on your calendar to commit to writing the proposal
- Unqualified enthusiasm to actually sit down and write the proposal

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GRANT WRITERS' SEMINARS AND WORKSHOPS

USE KNOWLEDGE OF THE REVIEW PROCESS TO INFORM WRITING THE REST OF YOUR PROPOSAL

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UNDERSTAND LINKAGE OF REVIEW CRITERIA TO COMPONENTS

REVIEW CRITERION	PROPOSAL COMPONENTS
SIGNIFICANCE	Significance, Broader Impacts
INVESTIGATOR(S)	Biographical Sketch / Resume
INNOVATION	Innovation / Originality
APPROACH	Plan of Work / Approach
ENVIRONMENT	Facilities, Equipment & Other Resources

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
Significance	X	X	X	X	X	X
Budget	X	X	X	X	X	X
Applicant/Env't	X	X	X	X	X	X
Background	X	X	X	X	X	X
Previous experience or preliminary data	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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PLAN OF WORK

The *purpose* of the Plan of Work / Research Design / Approach section is to describe how the proposed research will be carried out. Every funding agency, without exception, considers Approach to be an important review consideration.

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PLAN OF WORK

In this section of the application, you will tell the reviewers/funding agency precisely:

- What you propose to do (in sufficient detail) and why
- Exactly how you propose to do it
- What you expect to accomplish once what you propose to do has been completed
- What might go wrong and how you will fix it if something does go wrong

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PLAN OF WORK

The “Approach” (NIH, AFRI) or “Research Plan” (NSF) is where you will tell the reviewers in detail exactly what you plan to do. The simplest way to address this section is to develop a detailed outline (with page limitations) that allows you to address all of the important items reviewers will be looking for, and then rigidly stick to that outline.

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PLAN OF WORK - NIH

Reviewers will be evaluating *scientific rigor*, the strict application of the scientific method to ensure robust and unbiased experimental design, methodology, analysis, interpretation, and reporting of results. Reviewers will also be looking for sufficient clarity in what you have written – transparency – so that others will be able to reproduce and extend your study findings

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PLAN OF WORK

- Adhere to a particular format if required by the application guidelines, but use clear hierarchical organization
- If you have a *choice*:
 - Organize the plan of work by specific aim/objective
 - Organize the overall work plan as a single entity

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"PARALLEL ACTIVITY" FORMAT

1. Specific Aim/Goal #1
 - Introduction
 - Research Design
 - Expected Results/Outcomes
 - Potential Problems & Alternative Strategies
2. Specific Aim/Goal #2
 - Introduction
 - Research Design
 - Expected Results/Outcomes
 - Potential Problems & Alternative Strategies

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PLAN OF WORK

Each Specific Aim / Objective is a subsection:

- Write
- second ➤ Introduction
 - first {
 - Research Design
 - Expected Results/Outcomes
 - Potential Problems & Alternative Strategies

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"SINGLE ENTITY" FORMAT

- Introduction
- Research Design Statement
- Setting
- Participants (inclusion/exclusion criteria)
- Recruitment, Screening, Randomization
- Study Conditions (experimental & control conditions)
- Data Collection Procedures
- Variables and Measures
- Power Analysis and Sample Size Determination
- Analysis
 - Specific Aim #1
 - Specific Aim #2
- Expected Outcomes
- Potential Problems & Alternative Strategies

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PLAN OF WORK

RESEARCH DESIGN

- Use separate paragraphs/sections to develop each set of studies
- Avoid inclusion of mindless detail – i.e., what could be copied from a methods manual
- Succinctly provide only meaningful detail – what can't be found in a methods manual
- Refer to, don't detail, anything described in your team's peer-reviewed publications
- Express confidence in your ability to accomplish aims
- Don't forget to include power of study / statistics

GWSW*

PLAN OF WORK - **NIH**

RESEARCH DESIGN (Cont'd)

- If human subjects and/or vertebrate laboratory animals are proposed, design should include both sexes (unless only one can be justified)
- Design should allow disaggregation of data, so that sex-based comparisons can be made
- NIH **strongly encourages** investigators to discuss sex-based design and analysis with program staff before submitting a proposal

GWSW*

PLAN OF WORK - NIH

RESEARCH DESIGN (Cont'd)

- Other biological variables, such as weight, age and underlying health condition must be considered, as well
- The identity of cell lines and other biological and chemical resources must be authenticated
- See Appendix Three to this Handout for links to more information and FAQs regarding rigor of design and implementation, as well as transparency

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PLAN OF WORK

EXPECTED OUTCOMES

- Succinctly summarize what you expect your most important results will be
- This is the “endpoint” of all of the “we will do this and this” proposed in your work design
- “At the completion of this part of the project, we expect to have determined....”
- Emphasize how they will contribute to the achievement of your objective for the aim
- Results are important, but they can potentially be compromised by....

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PLAN OF WORK

POTENTIAL PROBLEMS & ALTERNATIVE STRATEGIES

- If potential problems exist, be certain to acknowledge them (You can be certain that reviewers will)
- First indicate why the problem is not likely
- Then offer alternative strategies --- *but*
- Do not overemphasize the problems (This is not intended to be self-incrimination)
- Write these as *conditionals* (They are only what you “might” or “would” do)

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PLAN OF WORK

POTENTIAL PROBLEMS?

- The working hypothesis might not test valid
- You may encounter difficulty in developing the necessary sensitivity for an assay
- You may not have *documented* experience with a non-standard technique
- Unforeseen problems in recruiting or retaining study subjects

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PLAN OF WORK

POTENTIAL PROBLEMS & ALTERNATIVE STRATEGIES

"In this aim, we have proposed to do
Based on the strength of our preliminary data, we expect this approach to be successful. Nevertheless, it is possible thatmight occur. In that unlikely event, we would Another problem that we might encounter is..... This seems unlikely because...."

AVOID SAME LANGUAGE EVERY TIME

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INCLUSION OF A TIMETABLE IS HELPFUL FOR REVIEWERS

- Include a timetable that reflects when stages of the project will be completed
- Use your specific aims/goals as main headings in the table
- Under each aim/goal list benchmarks that reflect the pace at which progress is expected to be made

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TIMETABLE

STUDY AIMS & BENCHMARKS	Yr 1	Yr 2	Yr 3	Yr 4
Study start up, hiring, approvals				
Specific Aim #1				
Completion of Experiment 1.1				
Completion of Experiment 1.2				
Completion of Experiment 1.3				
Specific Aim #2				
Completion of Experiment 2.1				
Completion of Experiment 2.2				
Manuscript preparation				

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PLAN OF WORK

Each Specific Aim / Objective is a subsection:

- Write
- second > Introduction
 - first {
 - > Research Design
 - > Expected Results/Outcomes
 - > Potential Problems & Alternative Strategies

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PLAN OF WORK

INTRODUCTION

A work plan should begin with an overview paragraph that includes: an *orientation focus* of the problem the aim will address, an identification of the *objective*, the *working hypothesis* (if applicable) to be tested, the *overall strategy* (or strategies) that will be used or activities to be carried out, the *rationale* for the work that will be proposed, and/or any *expected outcomes* that are anticipated. (6-8 sentences MAX)

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
Budget	X	X	X	X	X	X
Applicant/Env't	X	X	X	X	X	X
Background	X	X	X	X	X	X
Previous experience or preliminary data	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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SIGNIFICANCE

The *purpose* of the Significance section is to place your proposed work within the contextual framework of the overall mission of the funding agency from whom you will be requesting funding. (How would the funding agency be able to justify support of your project?) Every funding agency, without exception, considers Significance to be a key review consideration.

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SIGNIFICANCE

EXAMPLES

- NIH: First two subsections of the Research Strategy “Significance” and “Innovation”
- NSF: Section following the Overview and Objectives section “Expected Significance”; also “Broader Impacts”
- USDA/AFRI: Part Two of the Narrative in a section “Rationale and Significance”

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SIGNIFICANCE

SIGNIFICANCE PARAGRAPH(S):

- This section should preferably follow after the Overview section
- Make it easy for reviewers and the funding agency to identify the importance and impact of the work
- Significance projected must always be *directly relevant* to the mission of the funding agency (Reread mission statement)

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SIGNIFICANCE

SIGNIFICANCE PARAGRAPH(S):

Part 1: Substantiate, with documentation from the literature, there is a critical need, that it's an important problem/issue, and point out what your contribution is expected to be

Part 2: Italicized statement of significance: *This project is, therefore, significant because.....*

Part 3: List of benefits and impact on the field that can be credibly expected to accrue from the critical need having successfully been addressed

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SIGNIFICANCE - NIH

- The purpose for this subsection is to:
 - Substantiate the existence of the problem or critical barrier to progress in your field
 - Present and establish the *rigor* of prior research (others' and your own) supporting the project
 - Tell reviewers how/why your contribution will have an NIH-relevant positive impact

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NIH SIGNIFICANCE SUBSECTION: RIGOR OF PRIOR RESEARCH

- *Rigor* of the prior research refers to the key support upon which you are building your proposal, including the extent to which prior researchers have strictly applied the scientific method to ensure unbiased and well-controlled experimental design, methodology, analysis, interpretation and reporting of results

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SUGGESTED FORMAT FOR NIH SIGNIFICANCE SUBSECTION

SIGNIFICANCE

- Importance of the Problem to Be Addressed
- Rigor of the Prior Research Supporting the Aims
 - Aim #1 (Literature & Preliminary Results)
 - Aim #2 (Literature & Preliminary Results)
 - Aim #3 [if applicable] (Literature & Preliminary Results)
- Significance of the Expected Research Contribution

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INNOVATION - NIH

In the current NIH application format, applicants are *required* to include a separate subsection entitled "Innovation," which GWSW recommends should follow the Significance subsection and be 1/3 to 1/2 of a page. The NIH considers *Innovation* to be a key review criterion.

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THE DIFFERENCE BETWEEN SIGNIFICANCE & INNOVATION

- Significance: The positive effect that the successful completion of your research project is likely to have on addressing an important agency-relevant problem
- Innovation: A new and substantially different way of addressing an important, agency-relevant problem that requires departure from the status quo and could lead to new horizons

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WHAT CONSTITUTES INNOVATION?

- Does the application challenge or seek to shift current research or clinical paradigms?
- Does the research employ novel theoretical concepts, approaches, methodologies or instrumentation?
- Are any of the above novel to one particular field or novel in a broad sense?

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INNOVATION - **NIH**

Part 1: Document the existing strategies (i.e., *status quo*) being used to address a problem (or similar problems) and their limitations or why they have been unsatisfactory

Part 2: Italicized statement of innovation:

The proposed research is innovative, in our opinion, because.....

Part 3: Summarize the advancements that are likely to be possible with this new approach that would probably not have been possible without this innovative new approach

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INNOVATION

CREATIVITY, ORIGINALITY & TRANSFORMATIVE POTENTIAL:

- Innovation is also a key review consideration for other funders (e.g., NSF, USDA)
- Should be addressed in the last, “payoff” paragraph of the Overview section
- Make sure your claim is credible
- Be especially careful with your claim of *transformative* potential – don’t overreach

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NSF DEFINITION OF TRANSFORMATIVE RESEARCH

“Transformative research involves ideas, discoveries, or tools that *radically change our understanding of an important existing scientific or engineering concept or educational practice or leads to the creation of a new paradigm or field of science, engineering, or education.* Such research challenges current understanding or provides pathways to new frontiers.”

<https://new.nsf.gov/funding/learn/research-types/transformative-research>

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BROADER IMPACTS - **NSF**

Definition: the potential of the project to benefit society and contribute to the achievement of specific, *desired societal outcomes.*

<https://new.nsf.gov/funding/learn/broader-impacts>
<https://www.nsf.gov/pubs/2002/nsf022/bicexamples.pdf>
<https://researchinsociety.org/>

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BROADER IMPACTS - NSF

“DESIRED SOCIETAL OUTCOMES”

- Full participation of women, persons with disabilities, and underrepresented minorities in STEM fields
- Inclusion of individuals from geographic regions that are underrepresented in STEM disciplines
- Improved STEM education and educator development at any level
- Increased public scientific literacy and public engagement with science and technology
- Improved well-being of individuals and society in general

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BROADER IMPACTS - NSF

“DESIRED SOCIETAL OUTCOMES”

- Development of a diverse, globally competitive STEM workforce
- Increased partnerships between academia, industry, and others
- Improved national security
- Increased economic competitiveness of the United States
- Use of science and technology to inform public policy
- Enhanced infrastructure for research and education
Among others

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BROADER IMPACTS - NSF

- Broader Impacts criterion must be addressed in detail in Project Description and briefly in Project Summary
- Activities should be creative, original (and, if possible, *transformative*); i.e., they cannot come across as “canned” or “predictable”
- Choose 2 or 3 and devote the last 1-2 pages of the Project Description for this section
- Should include an evaluation/assessment plan, if applicable

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GUIDE TO CREATING A GOOD EVALUATION PLAN

*The 2010 User-Friendly Handbook
for Project Evaluation*

<https://www.informalscience.org/2010-user-friendly-handbook-project-evaluation>

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BROADER IMPACTS - NSF

- Can consist of:
 1. The research itself
Why don't more minority students study science?
 2. Activities directly related to the research
Recruit minority students to work on the project
 3. Activities supported by, but complementary to, the research
Conduct outreach program at the local high school level to interest minority students in science careers

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
Budget	X	X	X	X	X	X
Applicant/Env't	X	X	X	X	X	X
<i>Background</i>	X	X	X	X	X	X
Previous experience or preliminary data	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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BACKGROUND

The *purpose* of the Background/Literature Review section is to place your proposed work within the contextual framework of the work done by others. (How have earlier investigators helped to establish the intellectual platform from which your own work will now be launched?)

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BACKGROUND

EXAMPLES

- NIH: Included as part of “Rigor of the Prior Research” within “Significance” subsection
- NSF: Part of “Background” section under “Review of the Relevant Literature”; also “Relationship to Other Work in Progress”
- USDA/AFRI: Part of the “Introduction” section under “Review of the Relevant Literature”

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TIPS ON WRITING THE REVIEW OF LITERATURE

- Write this subsection *after* the Research Design part of the specific aim has been written
- Provide an up-to-date *critical* review (not a litany) of *relevant* literature
- Make certain each major point discussed allows a conclusion to be reached
- Include italicized sentences that tell reviewers how and why what you have just reviewed helps justify the need for what is being proposed
- Try to cite contributions of possible reviewers

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NSF RELATION TO OTHER WORK IN PROGRESS

- By the PI: complementary to or even synergistic with existing projects; no conflict will be created by funding the new application
- By Others: not duplicative or even similar to what is being done elsewhere; stress novelty and “catalytic impact”

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
Budget	X	X	X	X	X	X
Applicant/Env't	X	X	X	X	X	X
<i>Background</i>	X	X	X	X	X	X
<i>Previous experience or preliminary data</i>	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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PRELIMINARY RESULTS

The *purpose* of Preliminary Results / Work Done by You (as the Applicant) and/or work done by your co-investigators is to support the feasibility of the proposed work in your hands and to document that you have actually contributed already to the project (or have experience with related projects) and are not relying exclusively on the work of others.

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PRELIMINARY RESULTS

EXAMPLES

- NIH: Included as part of “Rigor of the Prior Research” within “Significance” subsection
- NSF: Part of “Background” section titled “Preliminary Studies”
- USDA/AFRI: Part of “Introduction” section under “Preliminary Data”

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USE PRELIMINARY STUDIES TO SUPPORT FEASIBILITY

PRELIMINARY STUDIES: TECHNICAL

- Regardless of agency, data presented should be carefully selected to support feasibility
- Data presentation should be as simple as possible
- Design each figure or table to convey a single point or idea
- Avoid inclusion of extraneous or irrelevant data
- Vary the style of presentation to make the data maximally appealing

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USE PRELIMINARY STUDIES TO SUPPORT FEASIBILITY - NIH

PRELIMINARY DATA: TECHNICAL (Cont'd)

- If human subjects and/or vertebrate laboratory animals will be used in the proposed research, to properly inform research design, preliminary data should be generated using both sexes, unless use of a single sex can be justified

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USE PRELIMINARY STUDIES TO SUPPORT FEASIBILITY

PRELIMINARY STUDIES: EDITORIAL

- Lead the reviewers through the data; don't make them do the work
- Place supporting figures / tables as close to where they are referred to in the text as possible
- Include italicized sentences that tell reviewers how data presented support feasibility
- Be certain that print in size-reduced figures / tables is legible
- Put methodology into figure legends / footnotes to tables, *not* in the text

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
Budget	X	X	X	X	X	X
<i>Applicant/Env't</i>	X	X	X	X	X	X
<i>Background</i>	X	X	X	X	X	X
<i>Previous experience or preliminary data</i>	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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APPLICANT / CO-APPLICANT(S)

The *purpose* of the Biographical Sketch information is to provide documentation that *all* of the required expertise necessary to successfully pursue the proposed project is available among the various personnel listed to participate in the project. Every funding agency considers *Applicant* to be a key review consideration.

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FORMAT OF THE **NIH** BIOGRAPHICAL SKETCH (5 p)

- Education/Training - In chronological order
- Personal Statement (Paragraph) - To justify role on the project, including *research support* and up to 4 publications or non-publication research products
- Positions, Scientific Appointments & Honors - In *reverse* chronological order
- Contributions to Science - Up to 5 earlier projects, with framing of problem, key findings, influence on field, specific project role, and up to 4 publications or non-publication research products. Provide URL to full publication list

<https://grants.nih.gov/grants/forms/biosketch.htm>

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NON-PUBLICATION RESEARCH PRODUCTS

Audio or video products
Conference proceedings
Patents
Data and research materials
Databases
Educational aids or curricula
Instruments or equipment
Models
Protocols
Software or network

GWSW*

NSF BIOGRAPHICAL SKETCH

An NSF-approved format for the biosketch is required for all proposals

- SciENCv: Science Experts Network Curriculum Vitae (*required effective October 23, 2023*)
- ~~NSF Fillable PDF~~

<https://new.nsf.gov/funding/senior-personnel-documents#biographical-sketch-Obd>

GWSW*

5/20/24 – FORMAT OF THE NSF BIOGRAPHICAL SKETCH (3-p)

- Identifying Information - name, ORCID iD (optional), position title (e.g., Assistant Professor)
- Organization and Location - name of primary organization and where organization is located
- Professional Preparation - in *reverse* chronological order
- Appointments and Positions - in *reverse* chronological order (academic, professional, institutional)
- Products
 - Up to five most closely related to proposed project
 - Up to five other significant products

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5/20/24 – SYNERGISTIC ACTIVITIES

“The Synergistic Activities section has been removed from the biographical sketch.

This information is to be submitted [as a separate document of *up to one page* that includes a list of up to five distinct examples of broader impact activities] by individuals designated as senior/key persons as part of the senior/key personnel documents in Research.gov.”

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COLLABORATORS AND OTHER AFFILIATIONS INFORMATION

- NSF requires the use of an Excel spreadsheet template (<https://new.nsf.gov/funding/senior-personnel-documents#collaborators-and-other-affiliations-2b3>) for identifying the Collaborators & Other Affiliations Information of each senior project personnel to provide information regarding collaborators and other affiliations
- Submission of information on postdoctoral scholar sponsors is no longer required

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CONTENT OF THE **USDA** BIOGRAPHICAL SKETCH

Limited to two (2) pages each in length

- Education and Training
- Honors and Awards
- Research and Professional Experience
- Research grants received
- Synergistic Activities

No Page limit

- All publications in refereed journals during the past four (4) years, including those in press
- Relevant non-refereed technical publications

Separate List of Current and Pending Support

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BIOGRAPHICAL SKETCH

OVERCOMING POTENTIAL DIFFICULTIES

- It is important to identify appropriate co-P.I.'s/co-investigators/collaborators/consultants
- Don't limit yourself to your own institution; but justify feasibility of the collaboration
- Where appropriate, get letters of support / collaboration

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HOW TO MAKE USE OF YOUR COLLEAGUES

NSF INSTRUCTIONS TO APPLICANTS

“Any substantial collaboration with individuals not included in the budget should be described and documented with a letter from each collaborator...”

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HOW TO MAKE USE OF YOUR COLLEAGUES

Write your own drafts of
letters of collaboration

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HOW TO MAKE USE OF YOUR COLLEAGUES

An Unhelpful Letter of Support

Dear Dr. Robinson:

I will be happy to serve as a consultant on
your research proposal to the National Institutes of
Health. Please let me know how I can be of help.

Sincerely yours,

Noah Lott, Ph.D.

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HOW TO MAKE USE OF YOUR COLLEAGUES

A Helpful Letter of Support

Dear John,

This correspondence confirms my agreement to
collaborate on your proposed NIH application. I believe
that you are addressing an important issue in the field of
cancer therapeutics, and our expertise with culturing
primary human pancreatic cancer cells would be an
ideal addition to your proposed studies. We would be
pleased to help you set this up in your lab whenever you
are ready. I am looking forward to our continued
collaboration. Warmest personal regards,

Noah

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NSF COLLABORATION LETTERS

OPTIONAL FORMAT

"If the proposal submitted by Dr. [insert the full name of the Principal Investigator] entitled [insert the proposal title] is selected for funding by NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal."
[PAPPG, Chapter II:D.2.i(iii)]

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RESOURCES AND ENV'T

The information included in the Facilities and Resources is what reviewers will primarily use to formulate the strengths and weaknesses sections in the "Environment" core review criteria. Every funding agency considers Environment to be a key review consideration.

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TIPS ON PREPARING THE RESOURCES SECTION

SPECIFIC COMPONENTS TO INCLUDE:

- Environment – Contribution to success (NIH- "narrative overview" of qualifications directly relevant to the proposed research)
- (Institutional Commitment to ESI) – Position, space, financial, personnel, time, intellectual
- Facilities – Laboratory, Animal, Clinical, Computer, Office and Other Resources (Including core facilities, machine shop)
- Major Equipment – (Only that specifically needed to pursue the project)

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
Descriptive title	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
<i>Budget</i>	X	X	X	X	X	X
<i>Applicant/Env't</i>	X	X	X	X	X	X
<i>Background</i>	X	X	X	X	X	X
<i>Previous experience or preliminary data</i>	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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REALITIES OF BUDGET REVIEW

- Budget should never drive the proposal
- Your request must be credible
- Once credibility is lost, the budgetary ax will cut even deeper
- In some funding agencies, budgets are important factors in the decision process
- Reviewers will not invariably cut something
- *Most cuts are the fault of the applicant, not the reviewer*
- Lack of adequate justification is the usual reason for cuts (NSF five pages; AFRI, NIH unlimited)

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REALITIES OF BUDGET REVIEW

The underlying secrets to any successful budget preparation are to base them on real costs that have accurately been determined and then to JUSTIFY all budgeted expenses exhaustively in the proposal

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NSF BUDGET POLICY GUIDELINES

"Inclusion of *voluntary committed cost sharing is prohibited* and Line M on the proposal budget [Mandatory Cost Sharing] will not be available for use by the proposer. *In order for NSF, and its reviewers, to assess the scope of a proposed project, all organizational resources [including personnel] necessary for, and available to a project, must be described in the Facilities, Equipment and Other Resources section of the proposal.*" [PAPPG, Chapter II:D.2.f(xii)]

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NSF BUDGET POLICY GUIDELINES

VOLUNTARY COMMITTED COST SHARING-INTERPRETATION

- All individuals listed on the budget under "personnel" (including the PIs) must have associated salary
- Individuals who will contribute effort, but will not receive salary, should *not* be listed in the budget
- Such individuals should be listed under "Facilities, Equipment and other Resources" but *without* information on salary or fringe benefits
- List person-month effort under "Current & Pending Support" section

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APPROACH TO BUDGET FOR NIH MODULAR APPLICATIONS

- \$25,000 direct cost increments up to \$250,000
- Provide budget justification for personnel (position, role, and calendar months). This section should be *extensively* discussed
- Not necessary to include salaries, travel, supplies or other expenses
- Include total cost estimate, not breakout, for consortium/contractual arrangements
- Other Support section provided "just-in-time"
- Applications over \$250,000/yr must include detailed initial and subsequent year budgets
- Does *not* obviate necessity to develop a real budget

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

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<i>Descriptive title</i>	X	X	X	X	X	X
Abstract/Summary	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
<i>Budget</i>	X	X	X	X	X	X
<i>Applicant/Env't</i>	X	X	X	X	X	X
<i>Background</i>	X	X	X	X	X	X
<i>Previous experience or preliminary data</i>	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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TIPS ON TITLES

- The importance of the title of the application should not be taken lightly
- It contributes to the funding agency's (and reviewers') first impression of your proposal
- Therefore, it must be informative and engender enthusiasm
- It may well influence the assignment of the reviewers

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TIPS ON TITLES

- Know whether or not there are restrictions on its length
- If so, know whether the restriction pertains to characters or characters and spaces
- Begin by listing words that convey what the proposal is essentially about
- Arrange words/abbreviations into a compelling, informative title that fits the space

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

Component	NIH	American Heart Assn.	USDA	NEH	NSF	FORD Foundation
<i>Descriptive title</i>	X	X	X	X	X	X
<i>Abstract/Summary</i>	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
<i>Budget</i>	X	X	X	X	X	X
<i>Applicant/Env't</i>	X	X	X	X	X	X
<i>Background</i>	X	X	X	X	X	X
<i>Previous experience or preliminary data</i>	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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GENERAL TIPS ON PROJECT SUMMARY/ABSTRACT

- It is probably *the* most important section during review, because it will be read by *all* reviewers, not just those assigned
- It must be written in plain English, because it must be interpretable by laypersons
- Write it last, *but not at the last minute*
- Use third person in this section *only*
- Do not use it to summarize past accomplishments or to review background material
- Should be a stand-alone section
- Becomes part of the public domain; protect what you don't want revealed

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NSF PROJECT SUMMARY

- Allowed one full page
- First Part: overview of objectives and approach for the research (~1/3p)
- Second Part: address Intellectual Merit of the project, especially expected significance or potential to advance knowledge (~1/3p)
- Third Part: address Broader Impacts of project to society and potential to achieve *desired societal outcomes* (~1/3p).

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NIFA PROJECT SUMMARY

- Limited to 250 words
- All PDs/PIs (PD/PI) and affiliated organizations should be listed in addition to the title of the project
- Should be description of the activity to be undertaken and focus on: overall project goal(s) and supporting objectives; plans to accomplish project goal(s); and relevance of the project to the goals of the program

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NIH PROJECT SUMMARY

- Allowed 30 lines of text
- Should be a self-contained description of the project and should contain a statement of objectives and methods to be employed
- Must include application's broad, long-term objectives, specific aims, and description of the research design and methods for achieving stated goals

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TIPS ON WRITING THE NIH PROJECT NARRATIVE

- Purpose: to establish project's *relevance to public health*
- Accordingly, begin first sentence with something like, "The proposed project is *relevant to public health* because ..."
- Limited to no more than 2-3 sentences
- We recommend that you include relevance to the mission of the Institute or Center you are targeting here, rather than in the Project Summary

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GRANT APPLICATIONS TO VERY DIFFERENT FUNDING AGENCIES ARE VERY SIMILAR!

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<i>Descriptive title</i>	X	X	X	X	X	X
<i>Abstract/Summary</i>	X	X	X	X	X	X
<i>Overview</i>	X	X	X	X	X	X
<i>Significance</i>	X	X	X	X	X	X
<i>Budget</i>	X	X	X	X	X	X
<i>Applicant/Env't</i>	X	X	X	X	X	X
<i>Background</i>	X	X	X	X	X	X
<i>Previous experience or preliminary data</i>	X	X	X	X	X	X
<i>Plan of work</i>	X	X	X	X	X	X

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GRANT WRITERS' SEMINARS AND WORKSHOPS

PREPARATION OF THE APPLICATION

~ Internal Review ~

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PRE-SUBMISSION REVIEW

This is an absolute necessity!

- Get review from knowledgeable colleagues
- Choose the right colleagues to review your completed application
- Do *not* restrict yourself only to colleagues in your field
- Give your colleagues time enough to help
- Do not ask for help while you're still making changes

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GRANT WRITERS' SEMINARS AND WORKSHOPS

THE REVIEW PROCESS

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THE REVIEW PROCESS

A full appreciation of the review process, including knowing the identity of those individuals who will be involved in the evaluation of your ideas, can significantly improve your chances of success.

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HELP WITH THE PROCESS IS OUT THERE!

Videos: *NIH Peer Review Revealed*

- 15-minute video of a mock review group meeting illustrating how NIH applications are reviewed

<https://www.youtube.com/watch?v=IzBhKeR6VIE>

- 5-minute video offering tips for applicants

<https://www.youtube.com/watch?v=IAOGtrOpM6Q>

<https://public.csr.nih.gov/NewsAndPolicy/PeerReviewVideos>

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COMMON ASSUMPTIONS ABOUT PEER REVIEW

- All reviewers will share the applicant's interest and enthusiasm for the proposal
- All reviewers will possess expertise pertinent to the applicant's proposal
- All reviewers will spend the time to read the applicant's proposal in detail
- All reviewers will be fair and impartial in judging the merits of the proposal

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COMMON ASSUMPTIONS ABOUT PEER REVIEW

NOT
NECESSARILY!

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THE REVIEW PROCESS

- Understand the process by which your application will be reviewed
- If possible to identify potential reviewers, take advantage of it
- If you are given the option to recommend reviewers, take it
- Understand who the reviewers really are
- Be compatible with all grant application requirements
- Understand the applicable review criteria

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THE REVIEW PROCESS

- At the NSF, Program Officers direct reviews and contribute to discussion of merit. Reviews often involve panel members (2-3) and/or several “outside” reviewers
- At the NIH, study section chair directs discussion and panel members (2-3) evaluate merit *without* input from Program Officers
- At NIFA, Program Leader and a Panel Manager select reviewers, chair discussions of merit, and prioritize proposals. Panel members (3-4) and (sometimes) ad hoc reviewers (up to 3) participate in review

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KEY POINT

“To succeed in peer review, you must win over the assigned reviewers. They act as your advocates in guiding the review panel’s discussion of your application. Write and organize your application so the primary reviewer can readily grasp and explain what you are proposing and advocate for your application.”

https://grants.nih.gov/grants/writing_application.htm

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IDENTIFY YOUR POTENTIAL REVIEWERS, IF POSSIBLE

Why is it important to know the members of the review panel?

- Knowing expertise of members can help make proposal understandable to them
- You may identify the lack of a required expertise needed to review your proposal
- You may identify a direct competitor on the panel
- You *never* want to overlook citing the work of someone from your field

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THE REVIEW PROCESS

FEDERAL AGENCY REVIEWS

- You are permitted to suggest the need for a specific scientific expertise
- You are allowed / invited, in some cases, to suggest the names of specific reviewers
- You are allowed to request the exclusion of specific reviewers (but provide a reason for your request)

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WHO ARE YOUR REVIEWERS?

They are:

- Accomplished
- Dedicated
- Knowledgeable
- Conscientious
- Fair

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WHO ARE YOUR REVIEWERS—REALLY?

They are *actually*:

- Overcommitted and overworked
- Very tired
- Underpaid for their efforts
- Inherently skeptical and overly critical
- Looking for the easiest way to get the job done *well*

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THE REVIEW PROCESS

4.AND THEN **REALLY** READ THE APPLICATION INSTRUCTIONS

- Read the correct instructions
- Read all of the instructions
- Read the current instructions

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THE REVIEW PROCESS

TWO SETS OF REVIEW CRITERIA

- INTRINSIC (Unwritten)
 - A reviewer's first impression of your application
- EXTRINSIC (Defined)
 - The evaluation criteria established by the funding agency

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HOW DO REVIEWERS REVIEW PROPOSALS?

WHAT DO REVIEWERS LOOK FOR FIRST?

- What's the title? Is it interesting?
- Who is the applicant?
- What's the applicant institution?
- What's the basic idea? Is it within my area of expertise?
- Is the application "reviewer friendly"?

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IS YOUR PROPOSAL "REVIEWER FRIENDLY"?

TIPS TO MAKE YOUR PROPOSAL "REVIEWER FRIENDLY"

- Use legible font size
- Leave spaces between paragraphs
- Use adequate (but not excessive) margins
- Make certain all figures are legible and have detailed figure legends
- Avoid excessive emphasized text
- Provide a clear organizational structure/outline
- Always remember why reviewers read proposals!

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CORE REVIEW CRITERIA FOR **USDA-NIFA (AFRI)**

SCIENTIFIC MERIT, e.g.:

- Novelty, innovation, originality
- Clarity and delineation of objectives
- Feasibility, as reflected in preliminary data

QUALIFICATIONS, FACILITIES AND MANAGEMENT, e.g.:

- Qualifications of the research team
- Awareness of previous and alternative approaches
- Research environment; competence in area
- Planning and administration of the project

PROJECT RELEVANCE

- Focus on a specific priority area

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USDA-NIFA (AFRI) SCORING SYSTEM

- During the meeting, the panelists discuss the proposal and reviews and arrive at a consensus ranking.
- Ranking categories are "outstanding," "high priority," "medium priority," "low priority," and "do not fund."
- Only proposals ranked in the first three categories may be considered for funding; those ranked in the latter two are ineligible for an award.

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NIH's CORE REVIEW CRITERIA

For due dates before January 25, 2025

SIGNIFICANCE
INVESTIGATOR(S)
INNOVATION
APPROACH
ENVIRONMENT

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NIH's REVIEW PROCEDURES

- Each application usually assigned to 3 reviewers
- Assigned reviewers score (1-9) each core-review criterion and award a preliminary impact score (1-9) based on a synthesis/integration, not average, of individual scores
- Each assigned reviewer writes a paragraph explaining the main factors that informed preliminary impact score
- Preliminary impact scores determine which applications will be discussed / not discussed during study section meeting

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NIH's REVIEW PROCEDURES

Cont'd:

- Secret-ballot vote on discussed applications to determine overall impact score
- Unassigned reviewers normally vote within the range of overall impact scores of assigned reviewers
- SRO averages overall impact scores and multiplies by 10 to arrive at a final overall impact score
- Percentile rank over three meetings often used to normalize study section variability
- Additional criteria (e.g., *mission relevance*, *portfolio balance*) used by Program Staff to break ties

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NIH's CORE REVIEW CRITERIA

<https://grants.nih.gov/policy/peer/simplifying-review.htm>

For due dates on/after January 25, 2025

Factor 1: Importance of the Research

- Significance and Innovation
- Scored 1-9

Factor 2: Rigor and Feasibility

- Approach
- Scored 1-9

Factor 3: Expertise and Resources

- Investigator(s) and Environment
- Considered in overall impact but not scored; assessed as appropriate or gaps identified; gaps require explanation

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NSF MERIT REVIEW PRINCIPLES

ALL NSF PROJECTS MUST:

1. be of highest quality, having the potential to advance, if not *transform*, frontiers of knowledge
2. contribute more broadly to achieving societal goals through: a) the research itself, b) activities directly related to the research projects, or c) activities supported by, but complementary to, the research project itself
3. include *outcomes* related to both major review criteria and be assessed/evaluated at either the individual- or aggregated-project level, as appropriate.

GWSW*

NSF DEFINITION OF TRANSFORMATIVE RESEARCH

"Transformative research involves ideas, discoveries, or tools that *radically change our understanding of an important existing scientific or engineering concept or educational practice or leads to the creation of a new paradigm or field of science, engineering, or education*. Such research challenges current understanding or provides pathways to new frontiers."

<https://new.nsf.gov/funding/learn/research-types/transformative-research>

GWSW*

NSF MERIT REVIEW PRINCIPLES

What Principle #3 means is that PIs need to be more “accountable for carrying out the activities described in the funded project” than they have been in the past. In other words, PIs must follow through with, and complete, their proposed Broader Impacts, as well as research, activities.

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GUIDE TO CREATING A GOOD EVALUATION PLAN

*The 2010 User-Friendly Handbook
for Project Evaluation*

<https://www.informalscience.org/2010-user-friendly-handbook-project-evaluation>

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NSF MERIT REVIEW CRITERIA

When evaluating NSF proposals, reviewers should consider:

- What the proposers want to do (Objective)
- Why they want to do it (Rationale)
- How they plan to do it (Approach)
- How they will know if they succeed (Assessments & Evaluations)
- What benefits would accrue if the project is successful (Expectations and Impact)

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THE TWO **NSF** MERIT REVIEW CRITERIA

DEFINITIONS:

1. Intellectual Merit: the potential of the project to advance scientific knowledge
2. Broader Impacts: the potential of the project to benefit society and contribute to the achievement of specific, *desired societal outcomes*.

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NSF REVIEW CONSIDERATIONS FOR BOTH MERIT REVIEW CRITERIA

1. Expected outcomes will (a) advance scientific knowledge *and* (b) benefit society/societal outcomes
2. Proposed studies are creative, original & potentially transformative
3. Proposed activities are well-reasoned, organized, based on a sound rationale, & include a strategy to assess success
4. The research team is well-qualified to conduct the proposed research
5. Institutional resources available are adequate to carry out the proposed research

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NSF SCORING SYSTEM

- Likely will involve two-three panel reviewers and several “outside” reviewers
- Each reviewer should comment on both Intellectual Merit and Broader Impact criteria
- Each reviewer will evaluate the total proposal as Excellent, Very Good, Good, Fair, or Poor
- Overall evaluation will reflect a consensus of the panel as either above criteria or as High, Medium, Low Priority, or Not Competitive

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WHAT DO REVIEWERS LOOK FOR LAST?

What singles out this grant application from all others under consideration?

WHY IS THIS GRANT PROPOSAL SPECIAL AND, THEREFORE, DESERVING OF SUPPORT?

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REVIEW SUMMARY: Moderate Enthusiasm

"The panel considers this to be *a good proposal* from a highly talented beginning assistant professor in an excellent research environment. While he presents some interesting preliminary findings that serve as the basis for a project deemed to be highly significant scientifically, reviewers agreed that many of the proposed studies *do not flow logically and/or appear overly ambitious*. Some of the studies proposed are imaginative and will yield clear-cut results; in other cases, the inclusion of *testable hypotheses* would increase confidence that the studies are appropriately focused. The Broader Impacts of the work are only *indirectly addressed* and appear superficial and not well thought out."

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GRANT WRITERS' SEMINARS AND WORKSHOPS

wishes you

***success**
in writing winning grant
proposals!*

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**GRANT WRITERS' SEMINARS
AND WORKSHOPS**

WE VALUE YOUR FEEDBACK

Please complete a very short survey at:

https://www.surveymonkey.com/r/UArk_WWGP_9-3-4-24



THANK YOU!

GWSW®

CREATE A BULLETED OUTLINE: SMOKING RELAPSE POSTPARTUM

Summary of Funding Opportunity: A Program Announcement (PA) co-sponsored by NIDA and NICHD has been issued encouraging research investigating health disparities among minority and underserved women of child-bearing age.

Applicant Details: You are a social scientist who has an interest in decreasing health disparities related to tobacco control. Your published data have established that nearly 90% of low-income mothers who quit smoking during pregnancy resume smoking within three months postpartum, compared to rates of 40-50% for more affluent mothers. Your preliminary data suggest that the disparity is attributable to three sources of stress and/or anxiety that are especially prevalent among postpartum, low-income women: work, new marital problems, and reduced self image. You decide to respond to this PA.

First (Introductory) Paragraph:

Opening Sentence: Although overall rates of cigarette smoking in the U.S. have decreased during the past 20 years, smoking rates have increased among low-income women of child-bearing age.

Current Knowledge: affirm existence of disparity:

- Your published data: 90% vs. 40-50% for affluent
- Complementary published data (refs)

Gap in Knowledge (What is Not Known): Underlying cause(s) of the disparity

Critical Need: To identify the underlying causes

Consequences of Not Meeting the Need: Unlikely that targeted intervention(s) can be developed without knowing the causes of disparity

Second (What, Why, Who?) Paragraph:

Long-Term Goal: Develop strategies to reduce tobacco-related morbidity and mortality

Overall Objective: To identify the underlying cause(s) for increased rate of smoking relapse among low-income women postpartum

Central Hypothesis and How Formulated: Attributable to psychosocial and behavioral factors

How Formulated: 1) Own preliminary data (see Research Strategy section for details) 2) Complementary published supporting data (cited but not described)

Rationale: Knowledge of underlying causes will inform development of interventions

Third (Specific Aims) Paragraph:

1. Psychosocial factors

Working Hypothesis: Loneliness, perceived infant irritability and/or decreased self image

2. Behavioral factors

Working Hypothesis: Not breastfeeding and physical inactivity

Fourth (Payoff) Paragraph:

Expected Outcomes: Will identify underlying causes for smoking-relapse disparity among postpartum, low-income women

Generality About Positive Impact:

- Vertically advance understanding of causes of smoking relapse
- Provide foundation for development of targeted interventions that will reduce/eliminate disparity

OVERVIEW / EXECUTIVE SUMMARY SECTION - Trade Routes to the East

Despite the recent development of trading with markets to the East, opportunities for individuals and governments in European countries to gain access to significant new products, as well as to grow economically, remain uncertain. Based upon the establishment of viable trade routes to the Orient, pioneered by the Italian Marco Polo and his colleagues, citizens of Europe are increasingly demanding the rich silk textiles, exotic teas, spices and other food products from the East. In this respect, such goods demand premier prices, and forward-thinking entrepreneurs and governments can be expected to accumulate significant wealth by capitalizing upon these opportunities to gain a dominant market share. Unfortunately, established trade routes to the East are currently extremely perilous and involve treacherous mountain passes, dangerous river crossings and threats from armed bandits. As a result, only ~15% of merchandise intended for European markets from the East actually reaches its destination. In addition, the risk to the lives of those involved in these trade routes is exceptionally high, and it is becoming increasingly difficult to recruit merchants willing to commit resources to such ventures. There exists, therefore, a critical need to identify alternative trade routes that can be used to cost-effectively and efficiently bring goods and merchandise from the East to markets in central Europe. In the absence of such alternative strategies, trade with the East will continue to be a challenging and economically risky practice that will impede Spain's economic growth.

Our long-term goal is to help establish Spain as the economic center of the European marketplace, thus bringing prosperity and economic wealth to both its citizens and the government of Spain. Our overall objective in this proposal is to identify and establish the viability of an alternative route for the efficient exchange of goods between Europe and the East. Our central hypothesis is that the earth is a sphere, and that ships sailing to the West will reach the Orient by an all-seas pathway, thus creating an economically viable, readily sustainable, and far superior trade route relative to existing overland trade routes through Eastern Europe and central Asia. We have based our central hypothesis upon the fact that virtually all celestial bodies observed to date appear round and, if the earth were also to be a celestial body, it should have a round/spherical form as well. In addition, and in contrast to "flat earth" proponents, ships sailing to the West beyond the visible horizon have not yet fallen off the edge. Finally, there exist earlier reports that the Norwegian explorer, Eric the Red, discovered in 997 A.D. a continent located three-months sail to the west of Iceland. We have speculated that this earlier identified land mass is associated with the far eastern lands currently reached by overland routes through Europe and eastern Asia. Our rationale for this project is that its successful completion would place Spain in a superior position to capitalize upon its extensive merchant marine inventory of galleons and trade ships to become an economic super-power in 15th century Europe. We are well-positioned to lead this project based on our 35 years of navigation experience, and we have recently created the first detailed map of the west coast of Iceland and Greenland. We plan to attain the overall objective by pursuing the following two specific aims:

Specific aim #1: *Identify an all-seas trade route with far eastern markets.* Our working hypothesis is that, by sailing due west-southwest from the shores of Spain, we will be able to reach far eastern trading communities within four months.

Specific aim #2: *Evaluate the economic viability of the new route.* Our working hypothesis is that we will be able to return to Spain within a year with our three galleons laden with exotic textiles, spices and foods from the East.

Upon the successful the completion of this project, it is our expectation that we will have established a novel, economically advantageous and safe trade route to eastern trading communities that will markedly enhance Spain's stature and economic well-being within the European economic community. We also expect that the successful completion of this project will have provided a tenfold return on Spain's initial investment in this proposed project. Not only would this accomplishment have an enormous positive impact on the financial security and stability of the government of Spain, it would also help to redefine existing concepts of our global environment, as well as establish novel new precedents for exploration and discovery.

Useful Resources on the Web

NIH

Rigor and Reproducibility in research

<http://www.nih.gov/science/reproducibility/index.htm>

<https://www.nih.gov/research-training/rigor-reproducibility/resources>

<https://grants.nih.gov/reproducibility/documents/grant-guideline.pdf>

<https://grants.nih.gov/grants/Rigor-and-Reproducibility-Chart-508.pdf>

Plan for Enhancing Diverse Perspectives

<https://braininitiative.nih.gov/vision/plan-enhancing-diverse-perspectives>

Reviewer guidelines

http://grants.nih.gov/grants/peer/reviewer_guidelines.htm (Scroll down to “Review Criteria Information” and “Scoring Information”)

<https://public.csr.nih.gov/ForReviewers>

Center for Scientific Review's resources for applicants

<https://public.csr.nih.gov/ForApplicants>

CSR's videos on peer review:

“What Happens to Your NIH Grant Application” <https://www.youtube.com/watch?v=Gg2nppTaLUw>

“Insider's Guide to NIH Peer Review for Applicants

<https://www.youtube.com/watch?v=DUNYjugBMXM&list=UUPe8bcUVSvkZhR5mfun0WoA&index=8>

NIH's “All About Grants” podcast episodes

<https://grants.nih.gov/news/virtual-learning/podcasts.htm>

Office of Extramural Research "About Grants"

https://grants.nih.gov/grants/about_grants.htm

General grant proposal writing strategy that is applicable across NIH Institutes and Centers, not just NIAID:

<https://www.niaid.nih.gov/grants-contracts/apply-grant>

NIH Vertebrate Animals

<https://olaw.nih.gov/guidance/vertebrate-animal-section.htm>

<https://olaw.nih.gov/vas-e-module/story.html>

NIAID's page of checklists for applications involving research with human subjects

<https://www.niaid.nih.gov/grants-contracts/checklists-human-subjects-grant-application>

Overview of new NIH policies on Human Subjects Research and Clinical Trials 15-minute video

https://grants.nih.gov/policy/clinical-trials/tutorial/story_html5.html?lms=1

Use of Cover Letters

A podcast covering when you should and shouldn't include a cover letter with your grant application.

<https://nexus.od.nih.gov/all/2018/04/16/cover-letters-and-their-appropriate-use-podcast-now-available/>

OER's page on Writing Your Application
http://grants.nih.gov/grants/writing_application.htm

Link to NIH listserv for Weekly "Table of Contents" with upcoming funding opportunities
<http://grants1.nih.gov/grants/guide/listserv.htm>

Understanding NRSA Fellowship "F" and Training "T" Grants: A Video Guide
<https://www.youtube.com/watch?v=qAGh4rsYK3E>

Data Management & Sharing Policy Overview
<https://sharing.nih.gov>

NIH-supported data repositories
<https://sharing.nih.gov/data-management-and-sharing-policy/sharing-scientific-data/repositories-for-sharing-scientific-data>

NSF:
"The 2010 User-Friendly Handbook for Project Evaluation"
<https://informal.science.org/wp-content/uploads/2019/02/TheUserFriendlyGuide.pdf>

Merit review page
https://www.nsf.gov/bfa/dias/policy/merit_review/

*NSF information on NSF Data Management Plan Requirements **by Directorate** (also, see link to FAQ at bottom of page)*
www.nsf.gov/bfa/dias/policy/dmp.jsp

DMPTool developed by a team of librarians
<https://cdlib.org/services/uc3/dmptool/>

A repository of data repositories
<http://www.re3data.org/>

"Note to Reviewers of CAREER Proposals"
http://www.nsf.gov/crssprgm/career/note_for_reviewers.pdf

Frequently Asked Questions – CAREER Awards
<https://www.nsf.gov/pubs/2022/nsf22100/nsf22100.jsp>

USDA
AFRI Request for Applications Resources (Including Part IV, C guidance)
<https://www.nifa.usda.gov/afri-request-applications-resources>

NIFA Peer Review Process
<https://nifa.usda.gov/resource/nifa-peer-review-process-competitive-grant-applications>

NIFA Electronic Application Information
<http://nifa.usda.gov/grants>

Guidelines for Reviewers
<http://nifa.usda.gov/resource/guidance-afri-reviewers>

USDA Data Management Plan info

<https://www.nifa.usda.gov/data-management-plan-nifa-funded-research-education-extension-projects>

USDA Logic Model Information

<https://nifa.usda.gov/resource/integrated-programs-logic-model-planning-process>

Interactive Logic Model Builder from the University of Minnesota – Requires setting up free account. Logic model output is not in conventional NIFA format, so some rearranging of the columns will be necessary.

Otherwise, a great tool!

<https://cyfar.org/node/176>

IES

Most recent general submission guide

<https://ies.ed.gov/funding/pdf/submissionguide.pdf> *IES funding page*

<http://ies.ed.gov/funding/>

Search Funded ED Grants

<http://ies.ed.gov/funding/grantsearch/index.asp>

IES Newsflash

<http://ies.ed.gov/newsflash/>

CDC

NIOSH extramural program

<https://www.cdc.gov/niosh/extramural-programs/php/index.html>

NEH

Grants Index

<https://www.neh.gov/grants>

Application Review Process

<http://www.neh.gov/grants/application-process>

HHMI

*Laboratory Set Up and Management—A **must** for new investigators – Chapters 3, 4, & 8*

<http://www.hhmi.org/developing-scientists/making-right-moves>

Full text article: <https://bmcpoc.biomedcentral.com/articles/10.1186/s12919-021-00214-7>

On Writing

Nature Magazine's English Communication for Scientists

<https://www.nature.com/scitable/ebooks/cntNm-14053993/contents/>

NIH's page on using clear language

<https://www.nih.gov/sites/default/files/institutes/plain-language/nih-plain-language-getting-started-brushing-up.pdf> *Illustrative plain language examples and revisions*

http://grants.nih.gov/grants/plain_language_examples.htm

Scientist as Visual Communicator

<http://www.printmag.com/design-inspiration/scientist-as-visual-communicator/>

About the Presenter:

John D. Robertson, Ph.D.

**Managing Member
Grant Writers' Seminars & Workshops**

Dr. Robertson received his Ph.D. from The University of Texas at Austin in pharmacology and toxicology in 1999. From 1999-2003, he was a postdoctoral research fellow at the Karolinska Institute in Stockholm, Sweden. He was subsequently hired in 2004 by the University of Kansas Medical Center in Kansas City, KS, where he was a faculty member in the Department of Pharmacology, Toxicology & Therapeutics, and an associate member of the University of Kansas Cancer Center. In 2010, he was recruited to teach grantsmanship by Grant Writers' Seminars and Workshops as an Associate Member. In 2017, he became the Managing Member of the company.

Dr. Robertson has been the recipient of competitive extramural funding from both the NIH and non-federal sources. He has authored 30 peer-reviewed journal articles and three book chapters. He has been a member of grant review panels, a reviewer for a number of biomedical journals, and served on editorial boards. In addition, he has been routinely recognized for excellence in teaching.

For more information about Grant Writers' Seminars & Workshops, please visit our website at www.grantcentral.com.